8. Execute and Display the Results for the Following Nmap Service Version and OS Detection Commands in Kali Linux Operating Systems

Note: Run these scans only against a system you own or have permission to test. For the examples below, 192.168.1.10 represents your authorized target IP address.

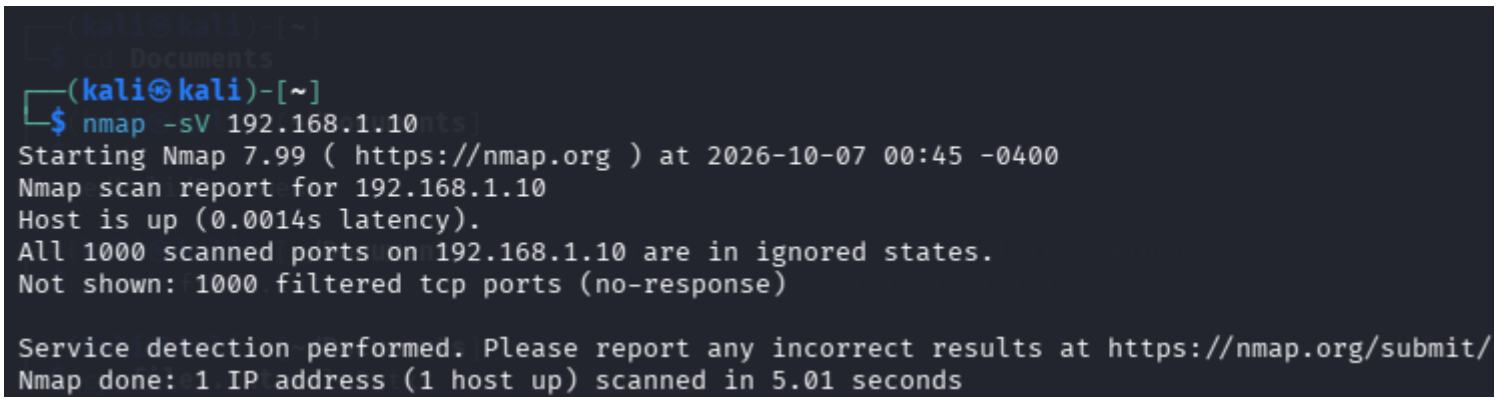
i) Detect the Version of Services Running

Use of Command:
The -sV option is used to **detect the versions of services** running on the target system.

Aim: To identify the services and their versions running on the target.

Command:

nmap -sV 192.168.1.10



Output:

Starting Nmap 7.95

Nmap scan report for 192.168.1.10

PORT STATE SERVICE VERSION

22/tcp open ssh OpenSSH 9.2

80/tcp open http Apache httpd 2.4.57

443/tcp open https Apache httpd 2.4.57

Nmap done: 1 IP address (1 host up) scanned

Result: The -sV option successfully detected the services and their versions running on the target.

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ii) Aggressive Scan

Use of Command:
The -A option performs an **aggressive scan**. It enables service/version detection, OS detection, script scanning, and traceroute.

Aim: To perform an aggressive scan and collect detailed information about the target.

Command:

nmap -A 192.168.1.10

```
┌──(kali㉿kali)-[~]
└─$ nmap -A 192.168.1.10
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 00:45 -0400
Nmap scan report for 192.168.1.10
Host is up (0.0031s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet
2%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-I
age HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband
2, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT     ADDRESS
1   2.96 ms 192.168.1.10

OS and Service detection performed. Please report any incorrect results at https://nmap.or
Nmap done: 1 IP address (1 host up) scanned in 12.95 seconds
```

Output:

Starting Nmap 7.95

Nmap scan report for 192.168.1.10

Host is up.

PORT STATE SERVICE VERSION

22/tcp open ssh OpenSSH 9.2

80/tcp open http Apache httpd 2.4.57

443/tcp open https Apache httpd 2.4.57

OS details: Linux

Service Info: OS: Linux

TRACEROUTE

HOP RTT ADDRESS

1 1.20 ms 192.168.1.10

Nmap done: 1 IP address (1 host up) scanned

Result: The -A option successfully performed an aggressive scan and provided detailed information about the target.

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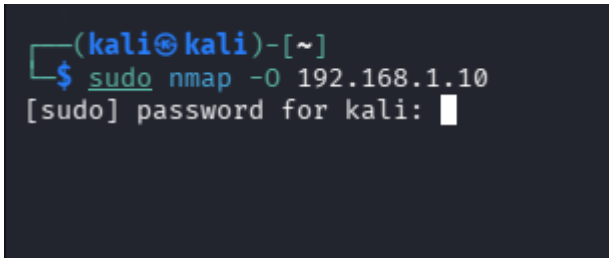
iii) Detect Operating System of the Target

Use of Command:
The -O option is used to **detect the operating system** of the target system.

Aim: To identify the operating system of the target.

Command:

sudo nmap -O 192.168.1.10



Output:

Starting Nmap 7.95

Nmap scan report for 192.168.1.10

Host is up.

PORT STATE SERVICE

22/tcp open ssh

80/tcp open http

Device type: general purpose

Running: Linux

OS details: Linux 5.x

Nmap done: 1 IP address (1 host up) scanned

Result: The -O option successfully detected the operating system of the target.